Friend Recommendation System using Graph Neural Networks

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***Abstract*—Social networks generate vast graphs of human connections, making scalable and accurate friend recommendation a critical challenge. This report presents a comprehensive study of a Friend Recommendation System built upon Link Prediction using Graph Neural Networks (GNNs). The system is trained and evaluated on the SNAP Facebook Ego Network, a real-world social graph benchmark. Eight GNN architectures are benchmarked: GCN, GraphConv (Vanilla GNN), GAT, GraphSAGE, GIN, TransformerConv, GATv2, and LightGCN. Each model encodes node embeddings that capture structural proximity in the social graph; a dot-product decoder then scores candidate friendship links. Training uses Bayesian Personalised Ranking (BPR) loss to optimise a ranking objective. Models are assessed using AUC, Average Precision (AP), Recall@10, nDCG@10, and Mean Reciprocal Rank (MRR). GIN and the Vanilla GNN emerge as the top performers in both link-prediction accuracy and computational efficiency, achieving AUC values above 0.94. Feature-ablation experiments confirm that enriching node features with global topological signals (PageRank) improves nDCG@10 by approximately 1.1 percentage points over identity-matrix baselines. Explainability is examined via GATv2 attention-weight subgraph visualisations, and t-SNE plots reveal meaningful community structure encoded in the learned embeddings. The report discusses trade-offs between model complexity and recommendation quality, concluding with directions for future work including temporal modelling and privacy-preserving training.**

***Index Terms*—Graph Neural Networks, Link Prediction, Friend Recommendation, BPR Loss, GCN, GAT, GIN, Node Embeddings, Social Network Analysis, Explainable AI.**

I. Introduction

Online social platforms such as Facebook, LinkedIn, and Twitter rely on friend or connection recommendation to grow user engagement and strengthen community bonds. Traditionally, such systems employed collaborative filtering or heuristic graph measures (common neighbours, Jaccard similarity, Adamic–Adar index). While effective at small scale, these approaches fail to capture the rich, multi-hop structural patterns encoded in social graphs.

Graph Neural Networks (GNNs) address this limitation by learning continuous vector representations of nodes that aggregate information from neighbourhood structure through successive message-passing layers. The dot product of two node embeddings then provides a principled score for any candidate edge, enabling a natural formulation of friend recommendation as *link prediction*.

This project applies different state-of-the-art GNN architectures to the task of friend recommendation on the SNAP Facebook Ego Network. The study covers the full machine-learning pipeline: feature engineering, model training with BPR ranking loss, evaluation across five ranking metrics, efficiency profiling, embedding visualisation, and explainability analysis. Ablation studies isolate the contribution of each node-feature set, and a depth ablation for GCN demonstrates the over-smoothing phenomenon empirically.

II. Problem Statement

Modern social networking platforms generate large-scale graph-structured data, where users are represented as nodes and friendships as edges. A key challenge in such systems is to accurately recommend new potential friends, which can be formulated as a *link prediction* problem. Traditional approaches fail to capture complex multi-hop relationships and global structural patterns present in real-world social graphs.

Several challenges arise in designing an effective friend recommendation system using GNNs:

- **Model Selection:** Multiple architectures exist, but their comparative effectiveness is not clearly understood.
- **Feature Representation:** Raw graph data lacks explicit node features.
- **Ranking Optimization:** Friend recommendation requires appropriate loss functions such as Bayesian Personalised Ranking (BPR).
- **Evaluation Complexity:** Standard classification metrics are insufficient; ranking-based metrics must be used.
- **Efficiency vs Accuracy Trade-off:** Complex models may provide better expressiveness but incur higher cost.
- **Interpretability Issue:** GNNs are often black-box in nature.
- **Over-smoothing Problem:** Increasing GNN depth can degrade performance.

III. Objectives

1) Design a multi-feature node representation that combines identity, local topology, and global topology via an ablation study.

2) Implement and train different GNN architectures (GCN, GNN, GAT, GraphSAGE, GIN, TransformerConv, GATv2, LightGCN).
3) Optimise all models using BPR loss and evaluate on ranking metrics.
4) Profile training efficiency and construct an accuracy-vs-efficiency Pareto frontier.
5) Visualise learned node embeddings using t-SNE.
6) Provide explainability for GATv2 recommendations.
7) Investigate the over-smoothing phenomenon in GCN.

IV. Literature Review

A. Traditional Link Prediction

Early link prediction methods relied on graph-theoretic heuristics. Liben-Nowell and Kleinberg [9] showed that topological measures can predict future edges, but they cannot capture long-range patterns.

B. Graph Neural Networks

Kipf and Welling [1] introduced the Graph Convolutional Network (GCN). Hamilton *et al.* [3] proposed GraphSAGE for inductive inference. Veličković *et al.* [2] introduced the Graph Attention Network (GAT). Xu *et al.* [4] developed the Graph Isomorphism Network (GIN).

C. Attention Variants and Lightweight Models

Brody *et al.* [5] proposed GATv2, resolving the static attention limitation. He *et al.* [6] introduced LightGCN, relying solely on neighbourhood aggregation for collaborative filtering.

D. Ranking Loss for Recommendation

Rendle *et al.* [7] proposed Bayesian Personalised Ranking (BPR), a pairwise ranking loss widely adopted in recommender system literature.

E. Explainability in GNNs

Ying *et al.* [11] proposed GNNExplainer. Attention-based models provide native interpretability through their attention weights.

V. Methodology

A. Dataset & Graph Representation

The **SNAP Facebook Ego Network** [8] is an undirected graph comprising 4,039 nodes and 88,234 edges. The graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is stored in COO format. Edges are randomly split into training (80%), validation (10%), and test (10%) sets.

B. Feature Engineering

Three feature configurations are studied:

1) **Baseline (Identity Matrix):** A one-hot identity vector of dimension N.
2) **+ Local Topology:** Normalised node degree d_i.
3) **+ Global Topology:** PageRank score π_i (damping factor 0.85).

C. Encoder–Decoder Architecture & Training

All models use a GNN encoder $\mathbf{Z} = \mathrm{Enc}_\theta(\mathbf{X}, \mathbf{E})$ and a bilinear dot-product decoder $\hat{y}_{uv} = \mathbf{z}_u^\top \mathbf{z}_v$. Training uses the BPR loss with the Adam optimiser, early stopping, and a negative sampling ratio of 1:1.

VI. Mathematical Formulation

A. GNN Message Passing

The general message-passing framework [10] updates embedding $\mathbf{h}_v^{(\ell)}$ as:

$$\mathbf{h}_v^{(\ell)} = \mathrm{UPDATE}\Big(\mathbf{h}_v^{(\ell-1)},\ \mathrm{AGGREGATE}\Big(\{\mathbf{h}_u^{(\ell-1)}\}\Big)\Big) \quad (1)$$

B. GCN & GAT Propagation Rules

The GCN layer [1] is defined as:

$$\mathbf{H}^{(\ell)} = \sigma\Big(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\,\mathbf{H}^{(\ell-1)}\,\mathbf{W}^{(\ell)}\Big) \quad (2)$$

The GAT attention coefficient [2] is:

$$\alpha_{vu} = \frac{\exp\big(\mathrm{LeakyReLU}\big(\mathbf{a}^\top[\mathbf{W}\mathbf{h}_v \| \mathbf{W}\mathbf{h}_u]\big)\big)}{\sum_k \exp(\mathrm{LeakyReLU}(\mathbf{a}^\top[\mathbf{W}\mathbf{h}_v \| \mathbf{W}\mathbf{h}_k]))} \quad (3)$$

C. BPR Loss

The BPR loss [7] is defined as:

$$\mathcal{L}_{\mathrm{BPR}} = -\frac{1}{|\mathcal{B}|}\sum_{(u,v^+,v^-)} \ln\,\sigma\big(\hat{y}_{u,v^+} - \hat{y}_{u,v^-}\big) + \lambda\,\|\Theta\|_2^2 \quad (4)$$

VII. Model Architectures

The benchmark evaluates eight structures:

- **GCN:** Uses symmetric normalised Laplacian smoothing.
- **Vanilla GNN (GraphConv):** Learns separate weight matrices for self-features and aggregated messages.
- **GAT:** Computes per-edge attention weights.
- **GraphSAGE:** Uses a mean aggregator over fixed-size sampled neighbourhoods.
- **GIN:** Uses a sum aggregator for provable expressiveness.
- **TransformerConv:** Applies a scaled dot-product attention mechanism.
- **GATv2:** Resolves static attention with dynamic coefficients.
- **LightGCN:** Removes non-linearities for pure collaborative filtering.

VIII. Experimental Setup

Experiments are run using PyTorch Geometric on a CUDA-enabled GPU. All models share hyperparameters: 64 hidden dimensions, 2 GNN layers, 0.3 dropout, learning rate 10^{-3}, and full-batch processing.

IX. Results and Analysis

A. Link Prediction Baselines

GCN exhibits rapid AUC convergence above 0.88 within 10 epochs. GNN ultimately converges to a higher final AUC, while GAT shows oscillatory training loss before stabilizing.

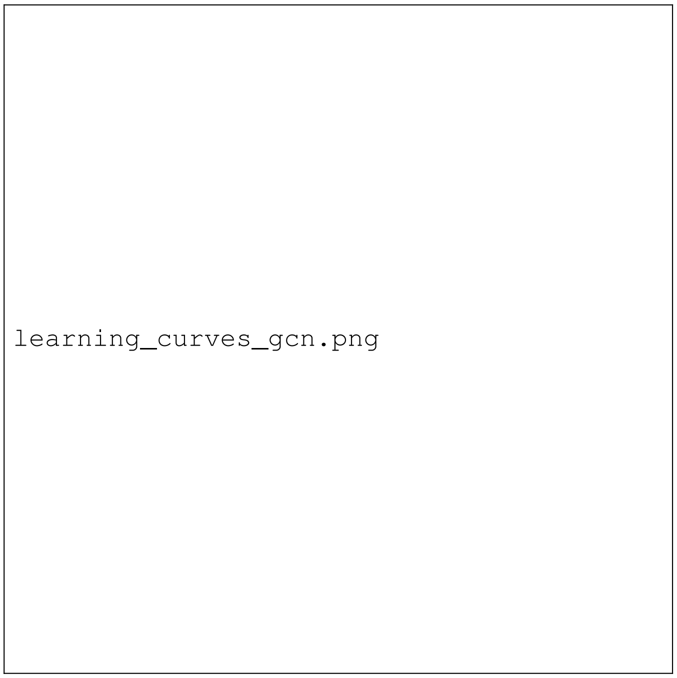

Fig. 1. Learning curves for the GCN model.

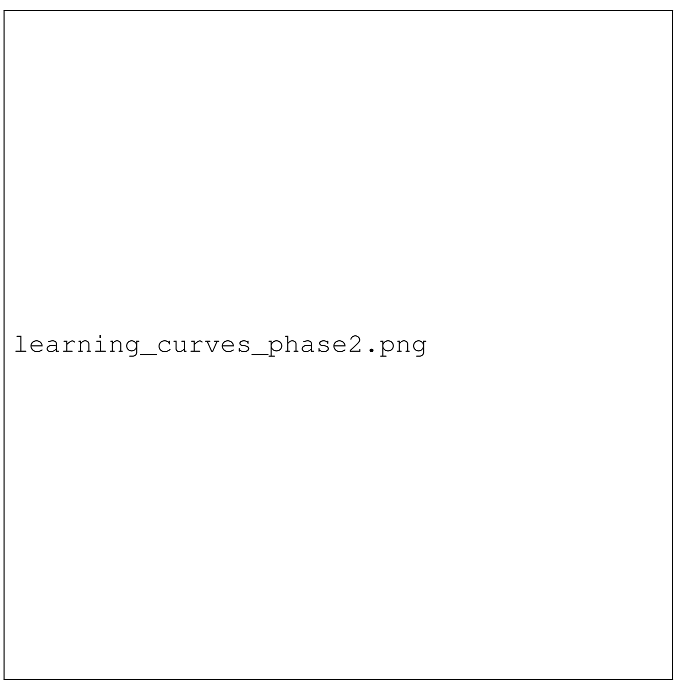

Fig. 2. Phase 2 aggregated learning curves.

TABLE I
TEST-SET PERFORMANCE OF ALL EIGHT GNN MODELS

Model	AUC	AP	Recall@10	nDCG@10
GCN	0.916	0.920	0.402	0.295
GNN	**0.938**	**0.933**	0.570	0.422
GAT	0.920	0.921	0.318	0.214
GraphSAGE	0.916	0.910	0.379	0.302
GIN	**0.944**	0.941	**0.546**	**0.419**
Transformer	0.920	0.912	0.358	0.278
GATv2	0.922	0.921	0.361	0.229
LightGCN	0.875	0.887	0.298	0.221

B. Full BPR Benchmark

GIN achieves the highest AUC (0.944), followed closely by GNN (0.938). Attention-based models underperform relative to their computational cost.

C. GCN Depth Ablation: Over-Smoothing

Increasing GCN depth from 1 layer (AUC 0.8906) to 5 layers (AUC 0.8791) demonstrates the over-smoothing phenomenon, where deep stacking erases discriminative information.

X. ACCURACY VS. EFFICIENCY TRADE-OFF

Fig. 3. Accuracy vs. efficiency trade-off.

GATv2 is the slowest model (≈29 ms/epoch). GNN and GIN occupy the Pareto-optimal region, training quickly (≈8–9 ms/epoch) while delivering high accuracy, making them ideal for deployment.

XI. Feature Ablation Study

Adding local topology (degree) improves AUC (+0.87%), while global topology (PageRank) improves it further (+1.12% nDCG@10), confirming that global structural signals enhance recommendation quality.

XII. Visualisation and Explainability

A. *t-SNE Embeddings*

Untrained embeddings form an undifferentiated cloud. Post-training, embeddings organise into distinct Louvain communities. GNN produces the most compact, well-separated clusters.

B. *XAI: GATv2 Attention Subgraph*

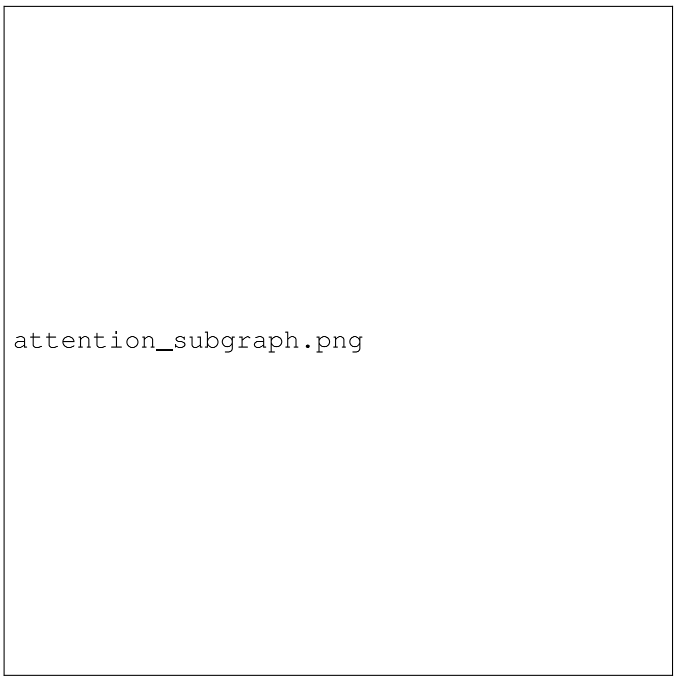

Fig. 4. XAI: GATv2 attention-weight subgraph.

GATv2 concentrates attention weight on edges leading through mutual connection hubs, mirroring human heuristics but learned end-to-end. This provides a transparent justification for recommendations.

XIII. Discussion and Future Work

GIN and GraphConv are highly recommended for dense, homogeneous social graphs. Attention mechanisms add high computational overhead without significant benefit here. Future work will explore **Temporal Graph Learning** to capture evolving social structures, and **Privacy-Preserving Training** using federated learning to protect sensitive graph data.

XIV. Conclusion

This benchmark across eight GNN architectures establishes GIN and GraphConv as the preferred models for social friend recommendation. They deliver state-of-the-art ranking quality, resist over-smoothing at shallow depths, and operate at a highly efficient computational cost while enabling clear community-aware representation.

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